

Heterogeneous Agent Model

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Recap: Motivation for Machine Learning

- ▶ Dynamic programming faces the curse of dimensionality
 - ▶ Parameterization problem
 - ▶ Classical functional forms (linear, polynomials, splines)
 - ▶ Number of basis terms grows rapidly with state dimension
 - ▶ Strong restrictions and poor capture of interactions
 - ▶ Grid / data problem
 - ▶ Grid-based DP requires N^d state points
 - ▶ Computation and memory grow exponentially
 - ▶ Sparse grids mitigate but do not eliminate the problem

How Machine Learning Addresses the Curse of Dimensionality

- ▶ Machine learning relaxes both constraints
 - ▶ Neural networks for parameterization
 - ▶ Flexible, high-dimensional function approximation
 - ▶ Captures nonlinearities and interactions
 - ▶ Stochastic simulation for data collection
 - ▶ Learn from simulated trajectories, not full grids
 - ▶ Scales to high-dimensional state spaces

RL for Economic Models: Data Generation

- ▶ One observation: (s, a, r, s')
 - ▶ s : taken as given
 - ▶ a : determined by policy $\sigma_\theta(s)$
 - ▶ r, s' : determined by the environment's reward function $R(s, a)$ and transition function $P(s' | s, a)$

- ▶ Policy Evaluation (Update Q_ϕ^σ) — requires σ_θ from Policy Improvement:

- ▶ Fixed-point iteration:

$$Q_\phi^\sigma(s, a) \leftarrow R(s, a) + \beta \sum_{s'} P(s' | s, a) Q_\phi^\sigma(s', \sigma_\theta(s'))$$

- ▶ Gradient-based update:

$$\phi \leftarrow \phi - \alpha \nabla_\phi \underbrace{\left(R(s, a) + \beta \sum_{s'} P(s' | s, a) Q_\phi^\sigma(s', \sigma_\theta(s')) - Q_\phi^\sigma(s, a) \right)^2}_{\text{Loss}}$$

RL for Economic Models: Q-Learning / Value Iteration

- ▶ One observation: (s, a, r, s')
- ▶ Q-Learning (Update Q_ϕ) — self-contained, only Q_ϕ needed:
 - ▶ Fixed-point iteration:

$$Q_\phi(s, a) \leftarrow R(s, a) + \beta \sum_{s'} P(s' | s, a) \max_{a'} Q_\phi(s', a')$$

- ▶ Gradient-based update:

$$\phi \leftarrow \phi - \alpha \nabla_\phi \left(\underbrace{R(s, a) + \beta \sum_{s'} P(s' | s, a) \max_{a'} Q_{\bar{\phi}}(s', a')}_{\text{Loss}} - Q_\phi(s, a) \right)^2$$

RL for Economic Models: Policy Improvement

- ▶ One observation: (s, a, r, s')
- ▶ Policy Improvement (Update σ_θ) — requires Q_ϕ from Policy Evaluation:
 - ▶ Fixed-point iteration:

$$\sigma_\theta(s) \leftarrow \arg \max_a Q_\phi(s, a)$$

- ▶ Gradient-based update:

$$\theta \leftarrow \theta + \alpha \underbrace{\nabla_\theta \log \sigma_\theta(a | s)}_{\text{Policy Gradient}} \cdot A(s, a)$$

where $A(s, a) = Q_\phi(s, a) - \mathbb{E}_{a \sim \sigma_\theta}[Q_\phi(s, a)]$ is the advantage

RL for Economic Models: Euler Equation Update

- ▶ One observation: (s, a, r, s') where $a = c$ (consumption), $s = (k, z)$
- ▶ Euler Equation Residual (Update c_θ) — self-contained, only c_θ needed:
 - ▶ Fixed-point iteration:

$$u'(c_\theta(s)) = \beta \sum_{s'} P(s' | s, a) [1 + r(s') - \delta] u'(c_\theta(s'))$$

- ▶ Gradient-based update:

$$\theta \leftarrow \theta - \alpha \nabla_\theta \underbrace{\left(u'(c_\theta(s)) - \beta \sum_{s'} P(s' | s) R'(s') u'(c_\theta(s')) \right)^2}_{\text{Euler Residual Loss}}$$

where $R'(s') = 1 + r(s') - \delta$ is the gross return

State Sampling Methods

- ▶ Updates require samples (s, a, r, s') ; given s , the tuple is determined by (σ_θ, R, P)
- ▶ Three approaches to obtain state s :

1. Grid-based (non-stochastic): Deterministic enumeration over $\mathcal{S}$

$$s \in \{s^{(1)}, s^{(2)}, \dots, s^{(N)}\}$$

2. Ergodic sampling: Draw from stationary distribution

$$s \sim \mu^* \quad (\text{long-run distribution under } \sigma^*)$$

3. Trajectory sampling: Sequential states from simulation

$$(s_0, a_0, r_0, s_1, a_1, r_1, s_2, \dots) \quad \text{where } s_{t+1} \sim P(\cdot \mid s_t, a_t)$$

State Sampling Methods: Comparison

	Grid-Based	Ergodic	Trajectory
Coverage	Full $\mathcal{S}$	Likely states	Path-dependent
Correlation	None	None (i.i.d.)	High (sequential)
Scalability	$O(N^d)$	Good	Good
Requirements	Grid construction	Estimate μ^*	Simulation

- ▶ Grid-based: Classical DP approach; curse of dimensionality
- ▶ Ergodic: Concentrates on economically relevant states; requires μ^* estimation
- ▶ Trajectory: Standard RL approach; requires decorrelation via replay buffer

Experience Replay Buffer

- ▶ Problem: data generated by sequential interaction is highly correlated
 - ▶ Violates i.i.d. assumptions underlying gradient-based optimization
 - ▶ Leads to unstable and inefficient learning
- ▶ Solution: store past transitions in a replay buffer $\mathcal{D} = \{(s, a, r, s')\}_{t=1}^N$
- ▶ Training loop:
 1. Interact with the environment using current policy π_θ (eval mode)
 2. Store observed transitions in $\mathcal{D}$
 3. Sample a random mini-batch from $\mathcal{D}$
 4. Update actor and critic using the sampled batch
- ▶ Key benefits:
 - ▶ Breaks temporal correlation and stabilizes training
 - ▶ Improves sample efficiency by reusing past experience

Online (On-Policy) vs. Offline (Off-Policy) Learning

- ▶ On-policy: Data generated by the current policy π_θ
 - ▶ Learning uses state distribution μ^{π_θ} ; policy and data evolve jointly
 - ▶ Advantages: Stable learning, no distribution mismatch
 - ▶ Limitations: Data discarded after each update; low sample efficiency
- ▶ Off-policy: Data collected by a different or past policy
 - ▶ Learning target decoupled from data collection
 - ▶ Advantages: High sample efficiency via replay; flexible data reuse
 - ▶ Limitations: Potential instability due to distribution mismatch
- ▶ Implications for economic models:
 - ▶ Simulation is cheap; training neural networks is expensive
 - ▶ Off-policy learning preferred to reuse simulated data efficiently

Parallel Simulation for Data Collection

- ▶ Vectorized environments: Simulate M agents in parallel

$$\{s_t^{(1)}, s_t^{(2)}, \dots, s_t^{(M)}\} \xrightarrow{\pi_\theta} \{a_t^{(1)}, a_t^{(2)}, \dots, a_t^{(M)}\}$$

- ▶ Each step generates M transitions simultaneously
- ▶ Benefits:
 - ▶ GPU parallelism for policy evaluation
 - ▶ Faster data collection
 - ▶ Better gradient estimates (more samples per update)

Heterogeneous Agent Models

- ▶ Representative agent models assume identical agents
 - ▶ Cannot capture wealth/income inequality
 - ▶ Miss redistributive effects of policy
 - ▶ Aggregate consumption = individual consumption (scaled)
- ▶ Heterogeneous agent models allow:
 - ▶ Idiosyncratic risk: individual-specific shocks (unemployment, health)
 - ▶ Incomplete markets: cannot fully insure against idiosyncratic risk
 - ▶ Endogenous wealth distribution: emerges from optimization
- ▶ Krusell & Smith (1998): First model combining HA with aggregate uncertainty

Krusell-Smith (1998): Key Innovation

- ▶ Challenge: With aggregate shocks, agents need to forecast future prices
- ▶ Future prices depend on future aggregate capital K' , which in turn depends on future wealth distribution μ'
- ▶ Future wealth distribution μ' depends on the current wealth distribution μ and the policy function σ
- ▶ The wealth distribution μ is infinite-dimensional
- ▶ Krusell-Smith insight:
 - ▶ Agents use bounded rationality
 - ▶ Approximate μ with a few moments (e.g., mean capital $\bar{K}$)
 - ▶ Empirically: first moment captures 99.9% of price variation

Model Environment: Preferences

- ▶ Continuum of agents indexed by $i \in [0, 1]$
- ▶ Time is discrete, infinite horizon: $t = 0, 1, 2, \dots$
- ▶ Preferences over consumption streams:

$$\mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t u(c_{it}) \right]$$

- ▶ Utility function (CRRA):

$$u(c) = \frac{c^{1-\gamma} - 1}{1-\gamma}, \quad \gamma > 0$$

- ▶ $\beta \in (0, 1)$: discount factor
- ▶ γ : coefficient of relative risk aversion

Model Environment: Labor Income Process

- ▶ Aggregate state $z \in \{z_g, z_b\}$ (good/bad times)
 - ▶ Follows Markov chain with transition matrix Π^z
 - ▶ Affects aggregate productivity
- ▶ Idiosyncratic state $\epsilon \in \{0, 1\}$ (unemployed/employed)
 - ▶ Transition probabilities depend on aggregate state z
 - ▶ $\pi(\epsilon' | \epsilon, z, z')$: probability of ϵ' given current (ϵ, z) and next z'
- ▶ Labor income:

$$y_{it} = \begin{cases} w_t \bar{\ell} & \text{if } \epsilon_{it} = 1 \text{ (employed)} \\ \mu^u & \text{if } \epsilon_{it} = 0 \text{ (unemployed, UI)} \end{cases}$$

where w_t is the wage and $\bar{\ell}$ is labor supply

Model Environment: Budget Constraint

- ▶ Agents can save in a single asset: capital $a_{it} \geq 0$
- ▶ Borrowing constraint: $a_{it} \geq \underline{a}$ (often $\underline{a} = 0$)
- ▶ Budget constraint:

$$c_{it} + a_{it+1} = (1 + r_t - \delta)a_{it} + y_{it}$$

where:

- ▶ r_t : rental rate of capital
 - ▶ δ : depreciation rate
 - ▶ y_{it} : labor income
- ▶ Incomplete markets: No insurance against ϵ shocks
- ▶ Agents self-insure through precautionary savings

Production Technology

- ▶ Representative firm with Cobb-Douglas technology:

$$Y_t = z_t K_t^\alpha L_t^{1-\alpha}$$

- ▶ $K_t = \int a_{it} di$: aggregate capital
- ▶ $L_t = \int \epsilon_{it} \bar{\ell} di$: aggregate effective labor
- ▶ Competitive factor markets imply:

$$r_t = \alpha z_t \left(\frac{K_t}{L_t} \right)^{\alpha-1}$$

$$w_t = (1 - \alpha) z_t \left(\frac{K_t}{L_t} \right)^\alpha$$

State Variables

- ▶ Individual state: (a, ϵ)
 - ▶ a : individual asset holdings
 - ▶ ϵ : employment status
- ▶ Aggregate state: (z, μ)
 - ▶ z : aggregate productivity shock
 - ▶ μ : distribution of agents over (a, ϵ)
- ▶ Key challenge: μ is infinite-dimensional
- ▶ Prices (r, w) are functions of (z, μ)

Household's Recursive Problem

- ▶ Value function $V(a, \epsilon; z, \mu)$ solves:

$$V(a, \epsilon; z, \mu) = \max_{c, a'} \{u(c) + \beta \mathbb{E} [V(a', \epsilon'; z', \mu') \mid \epsilon, z]\}$$

- ▶ Subject to:

$$c + a' = (1 + r(z, \mu) - \delta)a + y(\epsilon, w(z, \mu))$$

$$a' \geq \underline{a}$$

$$\mu' = \Gamma(z, \mu, z')$$

- ▶ Γ : law of motion for the distribution (unknown!)
- ▶ Policy function: $a' = g(a, \epsilon; z, \mu)$

Law of Motion for Distribution

- ▶ Given policy function g and transition π :

$$\mu'(A, \epsilon') = \int_{a \in A} \int_{\epsilon} \mathbf{1}\{g(a, \epsilon; z, \mu) \in A\} \cdot \pi(\epsilon' | \epsilon, z, z') d\mu(a, \epsilon)$$

- ▶ This is a functional equation in μ
- ▶ Aggregate capital evolves:

$$K' = \int a' d\mu' = \int g(a, \epsilon; z, \mu) d\mu(a, \epsilon)$$

Recursive Competitive Equilibrium: Definition

A recursive competitive equilibrium consists of:

1. Value function $V(a, \epsilon; z, \mu)$
2. Policy function $g(a, \epsilon; z, \mu)$
3. Price functions $r(z, \mu)$, $w(z, \mu)$
4. Law of motion $\Gamma(z, \mu, z')$

Such that:

- (i) Given prices and Γ , V and g solve household's problem
- (ii) Prices satisfy firm's FOCs:

$$r = \alpha z \left(\frac{K}{L} \right)^{\alpha-1}, \quad w = (1 - \alpha) z \left(\frac{K}{L} \right)^{\alpha}$$

- (iii) Markets clear: $K = \int a d\mu$
- (iv) Γ is consistent with g and transition probabilities

The Curse of Dimensionality

- ▶ Problem: Distribution μ is infinite-dimensional
- ▶ Cannot store μ on a computer directly
- ▶ Standard approaches fail:
 - ▶ Discretize a into N_a grid points
 - ▶ With $N_\epsilon = 2$ employment states
 - ▶ Need to track $N_a \times N_\epsilon$ probabilities
 - ▶ Value function becomes $V(a, \epsilon; z, \mu_1, \dots, \mu_{N_a \times N_\epsilon})$
- ▶ Even with $N_a = 100$: state space is $\mathbb{R}^{200}$
- ▶ Infeasible to solve by standard dynamic programming

Krusell-Smith's Bounded Rationality Approach

- ▶ Key insight: Agents don't know the true law of motion Γ
- ▶ Instead, agents use a perceived law of motion (PLM)
- ▶ Approximate μ with moments: $\vec{m} = (m_1, m_2, \dots, m_J)$
- ▶ Simplest case: $\vec{m} = \bar{K}$ (mean capital only)
- ▶ PLM for log aggregate capital:

$$\log K' = \begin{cases} a_0 + a_1 \log K & \text{if } z = z_g \\ b_0 + b_1 \log K & \text{if } z = z_b \end{cases}$$

- ▶ Coefficients (a_0, a_1, b_0, b_1) to be determined in equilibrium

Approximate Equilibrium

An approximate recursive competitive equilibrium consists of:

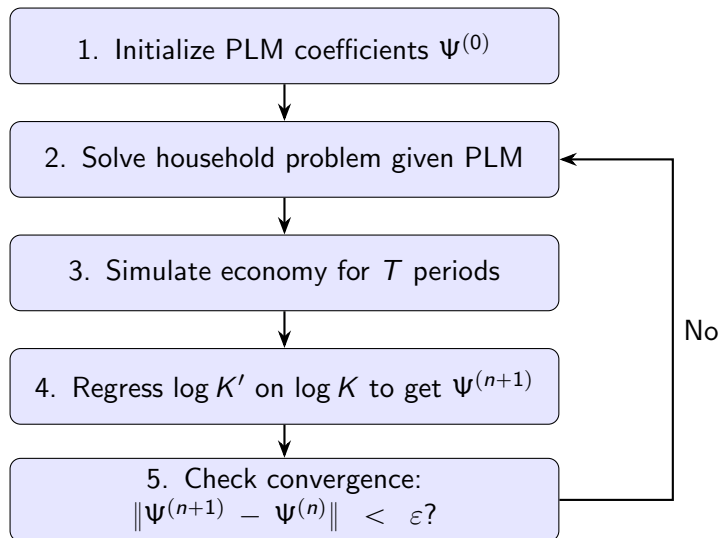
1. Value function $\tilde{V}(a, \epsilon; z, K)$
2. Policy function $\tilde{g}(a, \epsilon; z, K)$
3. Price functions $\tilde{r}(z, K)$, $\tilde{w}(z, K)$
4. PLM coefficients $\Psi = (a_0, a_1, b_0, b_1)$

Such that:

- (i) Given prices and PLM, $\tilde{V}$ and $\tilde{g}$ solve household's problem
- (ii) Prices from firm's FOCs with K as capital stock
- (iii) PLM Ψ provides a good forecast of actual K'

Accuracy criterion: $R^2 > 0.9999$ for forecasting K'

Krusell-Smith Algorithm: Overview



Step 1: Initialize PLM

- ▶ Set initial guess for PLM coefficients:

$$\psi^{(0)} = (a_0^{(0)}, a_1^{(0)}, b_0^{(0)}, b_1^{(0)})$$

- ▶ Common starting point:
 - ▶ $a_1^{(0)} = b_1^{(0)} \approx 0.99$ (high persistence)
 - ▶ $a_0^{(0)}, b_0^{(0)}$ chosen so steady-state K is consistent
- ▶ Or use steady-state Aiyagari model as initial guess

Step 2: Solve Household Problem

Given PLM Ψ , solve the Bellman equation:

$$\tilde{V}(a, \epsilon; z, K) = \max_{a' \geq \underline{a}} \left\{ u(c) + \beta \sum_{z', \epsilon'} \pi(z', \epsilon' | z, \epsilon) \tilde{V}(a', \epsilon'; z', K') \right\}$$

where:

$$\begin{aligned} c &= (1 + r(z, K) - \delta)a + y(\epsilon, w(z, K)) - a' \\ K' &= H(z, K; \Psi) \quad (\text{from PLM}) \end{aligned}$$

Solution methods:

- ▶ Value function iteration on grid
- ▶ Endogenous grid method (faster)
- ▶ Policy function iteration

Step 3: Simulate the Economy

- ▶ Track I agents for T periods (e.g., $I = 10,000$, $T = 11,000$)
- ▶ Initialize: draw (a_0^i, ϵ_0^i) from ergodic distribution
- ▶ For each period t :
 1. Compute aggregate capital: $K_t = \frac{1}{I} \sum_{i=1}^I a_t^i$
 2. Draw aggregate shock z_{t+1} from Markov chain
 3. For each agent i :
 - ▶ Draw ϵ_{t+1}^i from $\pi(\cdot | \epsilon_t^i, z_t, z_{t+1})$
 - ▶ Compute $a_{t+1}^i = \tilde{g}(a_t^i, \epsilon_t^i; z_t, K_t)$
- ▶ Discard first T_0 periods (burn-in, e.g., $T_0 = 1,000$)

Step 4: Update PLM via Regression

- ▶ From simulation, obtain time series $\{K_t, z_t\}_{t=T_0}^T$
- ▶ Run OLS regressions separately for each z :

$$\log K_{t+1} = a_0 + a_1 \log K_t + \varepsilon_t \quad \text{for } t : z_t = z_g$$

$$\log K_{t+1} = b_0 + b_1 \log K_t + \varepsilon_t \quad \text{for } t : z_t = z_b$$

- ▶ New PLM coefficients: $\Psi^{(n+1)} = (\hat{a}_0, \hat{a}_1, \hat{b}_0, \hat{b}_1)$
- ▶ Check R^2 for forecast quality (should be > 0.9999)

Step 5: Check Convergence

- ▶ Convergence criterion:

$$\|\Psi^{(n+1)} - \Psi^{(n)}\| < \varepsilon$$

- ▶ Typically $\varepsilon = 10^{-4}$ or smaller
- ▶ If not converged:
 - ▶ Update: $\Psi^{(n+1)} = \lambda \Psi^{(n+1)} + (1 - \lambda) \Psi^{(n)}$
 - ▶ Dampening factor $\lambda \in (0.3, 0.5)$ for stability
 - ▶ Return to Step 2
- ▶ If converged:
 - ▶ Verify $R^2 > 0.9999$ for both regressions
 - ▶ This validates the approximate equilibrium

The Near-Aggregation Property

- ▶ Surprising result: First moment ($\bar{K}$) is nearly sufficient
- ▶ $R^2 > 0.99999$ in original KS paper
- ▶ Intuition:
 - ▶ Employed agents (high income) save similarly
 - ▶ Unemployed agents (low income) dissave similarly
 - ▶ Individual differences average out in aggregation
 - ▶ Distribution shape matters little for aggregate K'
- ▶ This is called approximate aggregation
- ▶ Does NOT always hold (e.g., with larger shocks or more nonlinearity)